Activity 2: Process Module Architecture

Course: Fundamentals of Semiconductor Device Physics and Process Integration

Course code: TGS-2024049339

Duration: 40 minutes

TSC alignment: A1

PPT alignment: Trainer Slides v7.0, concept slides 120-258 and Activity 2 overview slide 442.

Workplace scenario

A fab is onboarding a mixed-experience team and needs a common architecture view that separates FEOL, MOL, BEOL and packaging loops.

Goal

Identify the major process loops, their purposes, inputs, outputs and control gates.

Files in this folder

- README.md / README.pdf - complete procedure and acceptance criteria.
- SCENARIO.md / SCENARIO.pdf - facts, assumptions and role brief.
- WORKSHEET.md / WORKSHEET.pdf - structured learner response.
- EVIDENCE-CHECKLIST.md / EVIDENCE-CHECKLIST.pdf - completion and review record.
- Any CSV file in this folder - supplied teaching data for analysis.

Step-by-step procedure

1. Sort the supplied process cards into FEOL, MOL, BEOL, wafer sort and assembly groups.
2. Arrange each group into a loop showing prepare, form, pattern, measure and release decisions.
3. Name the input and output of every process module.
4. Add the metrology or inspection gate that prevents a nonconforming wafer from advancing.
5. Annotate two feedback loops and explain what parameter each loop corrects.

Required evidence

Process architecture map with module boundaries, control gates and two feedback loops.

Include your completed worksheet, any calculations or annotated diagrams, the evidence checklist, and a short note identifying assumptions or unresolved evidence gaps.

Acceptance criteria

[ ] FEOL, MOL, BEOL and post-fab stages are separated

[ ] Every module has an input and output

[ ] At least two feedback loops are justified

Verification

Before submission, ask a peer to challenge one causal link. Revise the conclusion if the challenge reveals that evidence and inference have been mixed. Your final response must distinguish: observed fact, interpretation, decision, and follow-up check.

Troubleshooting

- If several mechanisms fit the symptom, choose a measurement that discriminates between them.
- If a process module lacks a defined input, output or control gate, the flow is incomplete.
- If an action is not tied to a verified cause, record it as a hypothesis test rather than corrective action.
- If calculations disagree, show the method and units so the discrepancy can be reviewed.